

Appendix B (categorical surprisal)

Model selection

Loading the data

```
# working directory
setwd("PATH")

# load data
data <- read.csv('3 - SPR main/analysis/readingtimes_88pp_residuals_manual-outliers-REVIEWS.csv', header = TRUE)
stiminfo <- read.csv('5 - metadata/frequency_length_position.csv', header = TRUE)

# set factors
data$agreement <- factor(data$agreement, levels=c('correct','incorrect'))
data$surprisal <- factor(data$surprisal, levels=c('low','high'))
data$correct_number <- factor(data$correct_number, levels=c("singular", "plural"))
data$set_number <- factor(data$set_number)

fmla <- 'logRTresidual ~ agreement * surprisal + agreement * correct_number +
        (1 + agreement * surprisal + agreement * correct_number | UserI
d) +
        (1 + agreement * surprisal | set_number )'

fixed <- 'logRTresidual ~ agreement * surprisal + agreement * correct_number'

fixed_effects <- list('logRTresidual ~ agreement * surprisal + agreement * correct_number', #
1
                        'logRTresidual ~ surprisal + agreement * correct_number',      #
2
                        'logRTresidual ~ agreement * correct_number',                  #
3
                        'logRTresidual ~ agreement * surprisal + correct_number',       #
4
                        'logRTresidual ~ agreement * surprisal',                       #
5
                        'logRTresidual ~ surprisal + agreement + correct_number',       #
6
                        'logRTresidual ~ agreement + correct_number',                  #
7
                        'logRTresidual ~ surprisal + correct_number',                  #
8
                        'logRTresidual ~ surprisal + agreement',                      #
9
                        'logRTresidual ~ surprisal',                                  #
10
                        'logRTresidual ~ agreement',                                  #
11
                        'logRTresidual ~ correct_number',                             #
12
                        'logRTresidual ~ ')
# 13

participant_effects <- c('(1 + agreement * surprisal + agreement * correct_number | UserId)',
                        '(1 + agreement * surprisal + correct_number | UserId)',
                        '(1 + agreement * correct_number + surprisal | UserId)',
                        '(1 + agreement + correct_number + surprisal | UserId)',
                        '(1 + agreement * correct_number | UserId)',
```

```

        '(1 + agreement + correct_number | UserId)',
        '(1 + agreement * surprisal | UserId)',
        '(1 + agreement + surprisal | UserId)',
        '(1 + surprisal + correct_number | UserId)',
        '(1 + agreement | UserId)',
        '(1 + surprisal | UserId)',
        '(1 + correct_number | UserId)',
        '(1 | UserId)'
    )

item_effects <- c('(1 + agreement * surprisal | set_number )',
                 '(1 + agreement + surprisal | set_number )',
                 '(1 + surprisal | set_number )',
                 '(1 + agreement | set_number )',
                 '(1 | set_number)')
)

# the random effects formulas
tmp_fm1as <- list()
for (randi in 1:length(item_effects)) {
  for (randp in 1:length(participant_effects)) {

    tmp_fm1a <- sprintf('%s + %s + %s', fixed, item_effects[[randi]], participant_effects[[r
andp]])
    tmp_fm1as <- c(tmp_fm1as, tmp_fm1a)

  }
}

```

Helper functions

```
n_permutations <- 1000
seed <- 42
set.seed(seed)

# formula
permute_lmer <- function(model_orig,
                          data,
                          n_permutations,
                          group_var = "UserId",
                          response_var = "logRTresidual",
                          REML=FALSE,
                          verbose=TRUE){

  formula <- formula(model_orig)
  coef_names <- names(fixef(model_orig))
  obs_coefs <- fixef(model_orig)

  # storage matrix (rows = perms, cols = fixed effects)
  perm_matrix <- matrix(NA, nrow = n_permutations, ncol = length(coef_names))
  colnames(perm_matrix) <- coef_names

  for (i in 1:n_permutations) {
    # shuffle labels
    data_perm <- data %>%
      group_by(.data[[group_var]]) %>%
      mutate(
        !!response_var := sample(.data[[response_var]])
      ) %>%
      ungroup()
    # Fit model to permuted data
    tryCatch({
      perm_model <- lmer(formula, data = data_perm, control=lmerControl(optimizer="nloptwra
p", optCtrl=list(maxfun=1e6)), REML = REML)
      perm_matrix[i, ] <- fixef(perm_model)
    }, error = function(e) {
      # Skip failed fits
    })

    if (verbose && i %% 50 == 0) {
      message(sprintf("Completed %d / %d permutations", i, n_permutations))
    }
  }

  perm_matrix <- perm_matrix[complete.cases(perm_matrix), , drop = FALSE]

  p_values <- sapply(seq_along(obs_coefs), function(j) {
    mean(abs(perm_matrix[, j]) >= abs(obs_coefs[j]))
  })
  names(p_values) <- coef_names

  return(list(
    observed_coefs = obs_coefs,
```

```

    permuted_coefs = perm_matrix,
    p_values = p_values
  ))
}

plot_permutation_densities <- function(result_list) {

  permuted_df <- as.data.frame(result_list$permuted_coefs)
  observed <- result_list$observed_coefs
  pvals <- result_list$p_values

  coef_names <- setdiff(names(permuted_df), "(Intercept)")

  plots <- lapply(coef_names, function(coef_name) {
    ggplot(permuted_df, aes(x = .data[[coef_name]])) +
      geom_density(fill = "skyblue", alpha = 0.5) +
      geom_vline(xintercept = observed[coef_name], color = "red", linewidth = 1) +
      labs(
        title = sprintf("%s\n(p = %.3f)", coef_name, pvals[coef_name]),
        x = "Coefficient value",
        y = "Density"
      ) +
      theme_minimal(base_size=10) +
      theme(plot.title = element_text(size = 10, hjust = 0.5)) # Center and shrink subplot titles
  })

  # Combine all plots into one figure
  combined_plot <- wrap_plots(plots, ncol = length(plots)) +
    plot_annotation(title = "Permutation Test Distributions for Fixed Effects",
      theme = theme(plot.title = element_text(hjust = 0.5, size = 14))
    )

  return(combined_plot)
}

```

```

count_random <- function mdl) {

  vc_list <- VarCorr(mdl)

  # Count variance components
  sum(sapply(vc_list, function(v) {
    n_std <- length(attr(v, "stddev"))
    n_corr <- if (!is.null(attr(v, "correlation"))) {
      length(attr(v, "correlation"))
    } else 0L
    n_std + n_corr
  })))
}

```

```

fit_random <- function(tmp_fm1a, data) {
  model <- lmer(tmp_fm1a, data, control=lmerControl(optimizer="nloptwrap", optCtrl=list(maxfun=2e5)), REML=FALSE)
}

fit_fixed <- function(tmp_fixed, random, data) {
  tmp_fm1a <- eval(parse(text=paste(tmp_fixed, " + ", random)))
  model <- lmer(tmp_fm1a, data, control=lmerControl(optimizer="nloptwrap", optCtrl=list(maxfun=1e6)), REML=FALSE)
}

```

Random effects - estimation of all models

Word index 5 (target)

```

models5 <- lapply(tmp_fm1as, fit_random, subset(data, word_index==5))
saveRDS(models5, file='PATH/analysis/models5CAT.Rdata')

```

Word index 6 (spill-over 1)

```

models6 <- lapply(tmp_fm1as, fit_random, subset(data, word_index==6))
saveRDS(models6, file='PATH/analysis/models6CAT.Rdata')

```

Word index 6 (spill-over 2)

```

models7 <- lapply(tmp_fm1as, fit_random, subset(data, word_index==7))
saveRDS(models7, file='PATH/analysis/models7CAT.Rdata')

```

Word index 6 (spill-over 3)

```

models8 <- lapply(tmp_fm1as, fit_random, subset(data, word_index==8))
saveRDS(models8, file='PATH/analysis/models8CAT.Rdata')

```

Evaluating the random models, fixed effects

estimation & model selection

Word index 5 (target)

```
models5 <- readRDS('PATH/analysis/models5CAT.Rdata')

good_models5 <- list()
for (mdli in 1:length(models5)) {
  mdl <- models5[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    good_models5 <- append(good_models5, mdl)
  }
}

# now we sort them for number of variables
# and their AIC -- we choose the model with the *highest number of variables* and *lowest AIC
* combination
rand_counts <- vapply(good_models5, count_random, integer(1))
AIC_vals <- sapply(good_models5, AIC)
nvar = order(-rand_counts)
aics = order(AIC_vals, decreasing=FALSE)

# now we take the sum of the position of each model
var_ix = match(1:length(good_models5), nvar)
aic_ix = match(1:length(good_models5), aics)

score <- var_ix + aic_ix

sorted_models5 <- good_models5[order(score)]

# we pick the largest one
start_model5 <- sorted_models5[[1]]
print(formula(start_model5))
```

```
## logRTresidual ~ agreement * surprisal + agreement * correct_number +
##      (1 + agreement * surprisal | set_number) + (1 + agreement |
##      UserId)
## <environment: 0x000001d0bda38400>
```

```

# now we reduce the fixed effects - STEP keeps reducing random effects too
# so we go back to manual selection
fixed5 <- lapply(fixed_effects, fit_fixed, "(1 + agreement * surprisal | set_number) + (1 + a
greement |
  UserId)", subset(data, word_index==5))

# save the fixed models
saveRDS(fixed5, file='PATH/analysis/fixed5CAT.Rdata')

final_models5 <- list()
for (mdli in 1:length(fixed5)) {
  mdl <- fixed5[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    final_models5 <- append(final_models5, mdl)
  }
}

fixed_aics = sapply(final_models5, AIC)
best_model5 = final_models5[order(fixed_aics, decreasing=FALSE)][[1]]
print(formula(best_model5))

```

```

## logRTresidual ~ agreement * correct_number + (1 + agreement *
##   surprisal | set_number) + (1 + agreement | UserId)
## <environment: 0x000001d0bc59e5e0>

```

```

summary(best_model5)

```



```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: tmp_fm1a
## Data: data
## Control: lmerControl(optimizer = "nloptwrap", optCtrl = list(maxfun = 1e+06))
##
##      AIC      BIC    loglik deviance df.resid
##    -89.8    21.0    62.9   -125.8    3447
##
## Scaled residuals:
##      Min      1Q  Median      3Q      Max
## -4.7864 -0.5699 -0.0888  0.4726  4.9896
##
## Random effects:
## Groups      Name                                Variance Std.Dev. Corr
## UserId      (Intercept)                        0.003571 0.05976
##              agreementincorrect                 0.002209 0.04699 -0.56
## set_number  (Intercept)                        0.002510 0.05010
##              agreementincorrect                 0.003014 0.05490  0.81
##              surprisalhigh                     0.004447 0.06669 -0.45 -0.16
##              agreementincorrect:surprisalhigh 0.013147 0.11466 -0.34 -0.82
## Residual                                         0.052206 0.22849
##
##
##
##
##
##
## -0.30
##
## Number of obs: 3465, groups:  UserId, 88; set_number, 40
##
## Fixed effects:
##
##              Estimate Std. Error    df t value
## (Intercept)    -0.08244    0.01393 62.87231  -5.918
## agreementincorrect    0.04743    0.01397 80.24726   3.395
## correct_numberplural    0.04243    0.01708 43.70037   2.484
## agreementincorrect:correct_numberplural -0.07447    0.01798 74.40522  -4.142
##
##              Pr(>|t|)
## (Intercept)    1.47e-07 ***
## agreementincorrect    0.00107 **
## correct_numberplural    0.01690 *
## agreementincorrect:correct_numberplural 9.00e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) agrmnt crct_
## agrmntncrrc -0.279
## crct_nmbrp -0.645  0.153
## agrmntncr:_  0.146 -0.677 -0.225
```

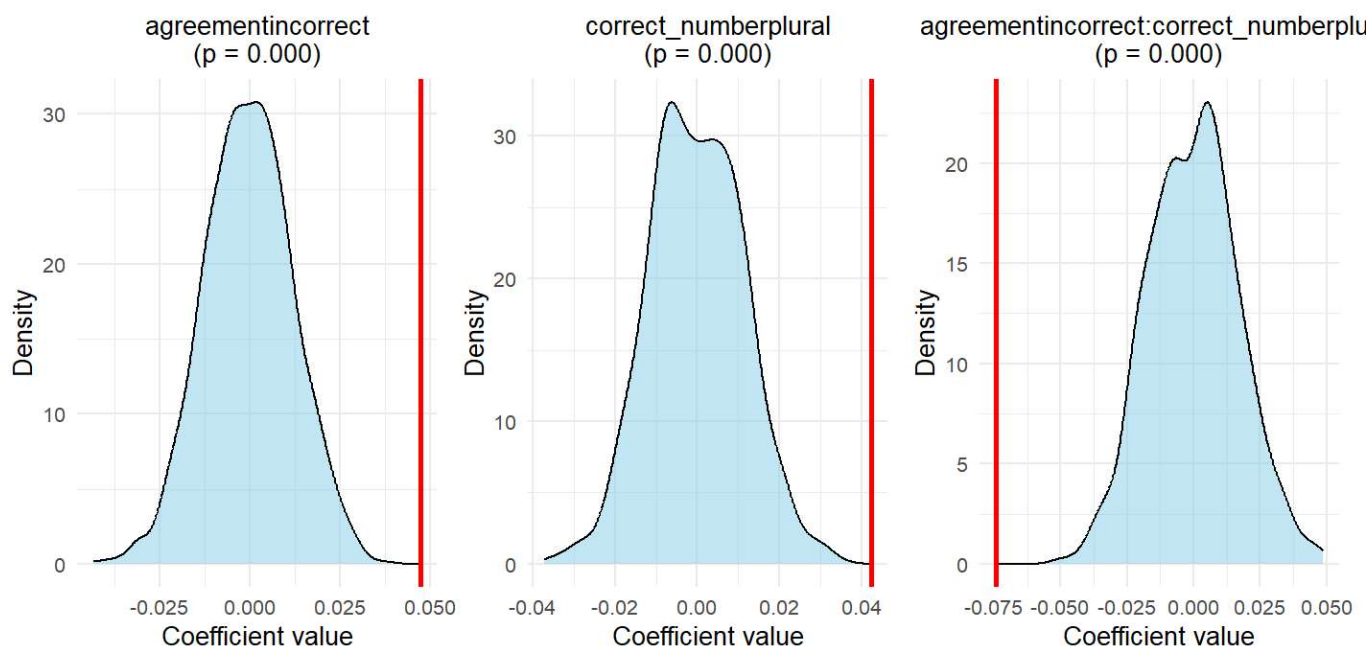
```
saveRDS(best_model5, file='PATH/analysis/best_model5CAT.Rdata')
```

Permutation test

```
best_model5 <- readRDS('PATH/analysis/best_model5CAT.Rdata')
perm5 <- permute_lmer(best_model5,
  subset(data, word_index==5),
  n_permutations,
  verbose=TRUE)
saveRDS(perm5, file='PATH/analysis/perm55CAT.Rdata')
```

```
plots5 <- plot_permutation_densities(perm5)
print(plots5)
```

Permutation Test Distributions for Fixed Effects



```
print(perm5$p_values)
```

```
##                (Intercept)                agreementincorrect
##                      0                      0
##      correct_numberplural agreementincorrect:correct_numberplural
##                      0                      0
```

Simple effects

```

# agreement by number
mdl5_agr_by_num <- emmeans(best_model5, specs='agreement', by='correct_number')

# number by agreement
mdl5_num_by_agr <- emmeans(best_model5, specs='correct_number', by='agreement')

# get results without adjustment
mdl5_agr_by_num_raw <- summary(pairs(mdl5_agr_by_num), adjust='none')
mdl5_num_by_agr_raw <- summary(pairs(mdl5_num_by_agr), adjust='none')

# extract pvalues
all_pvals <- c(mdl5_agr_by_num_raw$p.value, mdl5_num_by_agr_raw$p.value)
adj_pvals <- p.adjust(all_pvals, method='bonferroni')

# extract pvalues
mdl5_res_pair <- rbind(mdl5_agr_by_num_raw, mdl5_num_by_agr_raw)
mdl5_res_pair$adj.p.value <- adj_pvals
summary(mdl5_res_pair)

```

```

## correct_number agreement contrast      estimate    SE   df t.ratio
## singular      .      correct - incorrect -0.0474 0.0140 80.2  -3.395
## plural        .      correct - incorrect  0.0270 0.0133 78.6   2.026
## .             correct  singular - plural  -0.0424 0.0171 43.7  -2.484
## .             incorrect singular - plural  0.0320 0.0218 39.7   1.467
## p.value adj.p.value
## 0.0011    0.00428
## 0.0462    0.18465
## 0.0169    0.06760
## 0.1503    0.60120
##
## Degrees-of-freedom method: satterthwaite

```

Word index 6 (spill-over 1)

```
models6 <- readRDS('PATH/analysis/models6CAT.Rdata')

good_models6 <- list()
for (mdli in 1:length(models6)) {
  mdl <- models6[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code)))) {
    good_models6 <- append(good_models6, mdl)
  }
}

# now we sort them
rand_counts <- vapply(good_models6, count_random, integer(1))
AIC_vals <- sapply(good_models6, AIC)
nvar = order(-rand_counts)
aics = order(AIC_vals, decreasing=FALSE)

# now we take the sum of the position of each model
var_ix = match(1:length(good_models6), nvar)
aic_ix = match(1:length(good_models6), aics)
score <- var_ix + aic_ix

sorted_models6 <- good_models6[order(score)]

# we pick the best/largest one
start_model6 <- sorted_models6[[1]]
print(formula(start_model6))

## logRTresidual ~ agreement * surprisal + agreement * correct_number +
##      (1 + surprisal | set_number) + (1 + agreement | UserId)
## <environment: 0x000001d0bbdabe28>
```

```

# now we reduce the fixed effects - STEP keeps reducing random effects too
# so we go back to manual selection
fixed6 <- lapply(fixed_effects, fit_fixed, "(1 + surprisal | set_number) + (1 + agreement | U
serId)", subset(data, word_index==6))

# save the fixed models
saveRDS(fixed6, file='PATH/analysis/fixed6CAT.Rdata')

# now we check whether they converged
final_models6 <- list()
for (mdli in 1:length(fixed6)) {

  mdl <- fixed6[[mdli]]

  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code))) {
    final_models6 <- append(final_models6, mdl)
  }
}

# they did, so we pick the model with the highest AIC
fixed_aics = sapply(final_models6, AIC)
best_model6 = final_models6[order(fixed_aics, decreasing=FALSE)][[1]]
print(formula(best_model6))

```

```

## logRTresidual ~ agreement * correct_number + (1 + surprisal |
##      set_number) + (1 + agreement | UserId)
## <environment: 0x000001d0b439af60>

```

```
summary(best_model6)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: tmp_fm1a
## Data: data
## Control: lmerControl(optimizer = "nloptwrap", optCtrl = list(maxfun = 1e+06))
##
##      AIC      BIC    logLik deviance df.resid
##    202.1    269.7    -90.1    180.1     3437
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.1411 -0.6089 -0.0543  0.4944  5.2588
##
## Random effects:
## Groups      Name                Variance Std.Dev. Corr
## UserId      (Intercept)          0.001030 0.03209
##              agreementincorrect 0.005008 0.07077  0.67
## set_number  (Intercept)          0.005245 0.07242
##              surprisalhigh       0.002075 0.04555 -0.86
## Residual                                0.057936 0.24070
## Number of obs: 3448, groups:  UserId, 88; set_number, 40
##
## Fixed effects:
##                                     Estimate Std. Error      df
## (Intercept)                       -0.03749    0.01413    64.90033
## agreementincorrect                   0.08980    0.01410   220.62428
## correct_numberplural                 0.03043    0.01892    59.48713
## agreementincorrect:correct_numberplural -0.05043    0.01645  3206.87490
##                                     t value Pr(>|t|)
## (Intercept)                       -2.653  0.01002 *
## agreementincorrect                   6.370 1.08e-09 ***
## correct_numberplural                 1.608  0.11303
## agreementincorrect:correct_numberplural -3.066  0.00218 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) agrmnt crrcr_
## agrmntncrrc -0.265
## crrcr_nmbrp -0.703  0.263
## agrmntncr:_  0.301 -0.612 -0.429
```

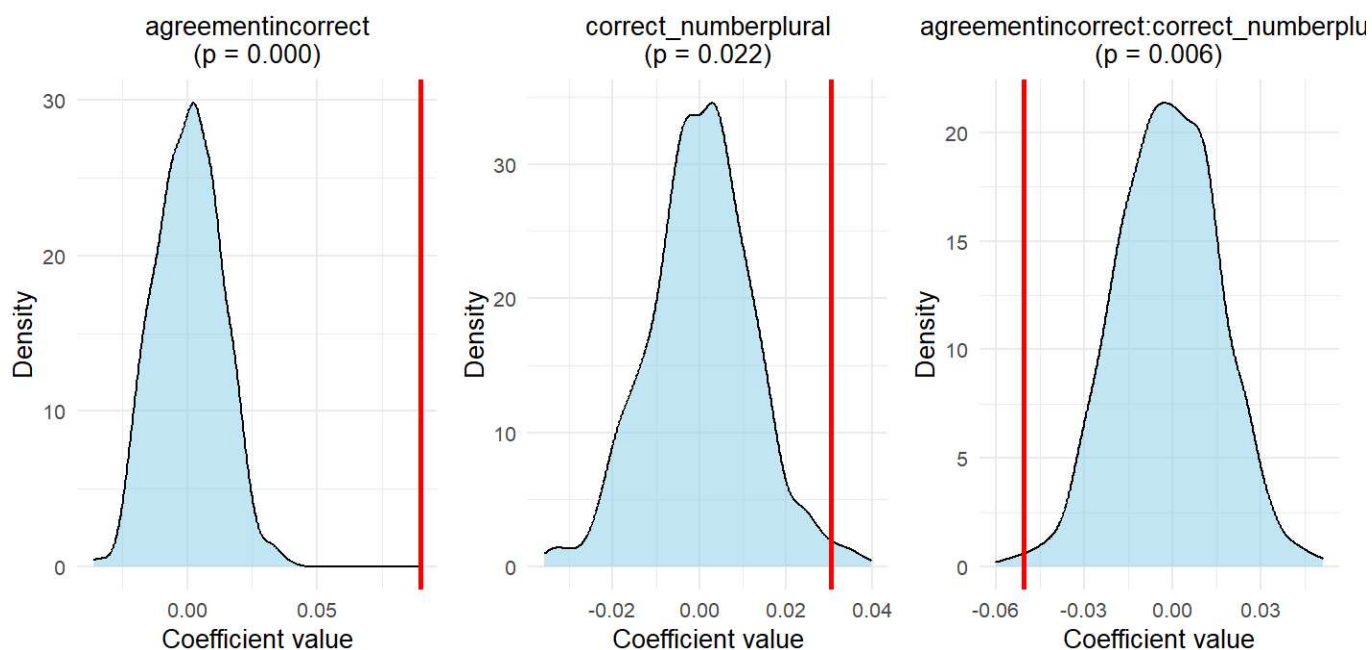
```
saveRDS(best_model6, file='PATH/analysis/best_model6CAT.Rdata')
```

Permutation test

```
# Load the best model
best_model6 <- readRDS('PATH/analysis/best_model6CAT.Rdata')
perm6 <- permute_lmer(best_model6,
                      subset(data, word_index==6),
                      n_permutations,
                      verbose=TRUE)
saveRDS(perm6, file='PATH/analysis/perm6CAT.Rdata')
```

```
plots6 <- plot_permutation_densities(perm6)
print(plots6)
```

Permutation Test Distributions for Fixed Effects



```
print(perm6$p_values)
```

```
##                (Intercept)                agreementincorrect
##                0.0000                0.0000
##      correct_numberplural agreementincorrect:correct_numberplural
##                0.022                0.006
```

Simple effects

```
# agreement by number
mdl6_agr_by_num <- emmeans(best_model6, specs='agreement', by='correct_number')

# number by agreement
mdl6_num_by_agr <- emmeans(best_model6, specs='correct_number', by='agreement')

# get results without adjustment
mdl6_agr_by_num_raw <- summary(pairs(mdl6_agr_by_num), adjust='none')
mdl6_num_by_agr_raw <- summary(pairs(mdl6_num_by_agr), adjust='none')

# extract pvalues
all_pvals <- c(mdl6_agr_by_num_raw$p.value, mdl6_num_by_agr_raw$p.value)
adj_pvals <- p.adjust(all_pvals, method='bonferroni')

# extract pvalues
mdl6_res_pair <- rbind(mdl6_agr_by_num_raw, mdl6_num_by_agr_raw)
mdl6_res_pair$adj.p.value <- adj_pvals
summary(mdl6_res_pair)
```

```
## correct_number agreement contrast      estimate      SE      df t.ratio
## singular      .      correct - incorrect -0.0898 0.0141 220.6 -6.370
## plural      .      correct - incorrect -0.0394 0.0136 192.7 -2.891
## .      correct      singular - plural -0.0304 0.0189  59.5 -1.608
## .      incorrect    singular - plural  0.0200 0.0190  60.6  1.052
## p.value adj.p.value
## <.0001      0.0000
## 0.0043      0.0171
## 0.1130      0.4521
## 0.2969      1.0000
##
## Degrees-of-freedom method: satterthwaite
```

Word index 7 (spill-over 2)

```
models7 <- readRDS('PATH/analysis/models7CAT.Rdata')

good_models7 <- list()
for (mdli in 1:length(models7)) {
  mdl <- models7[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code)))) {
    good_models7 <- append(good_models7, mdl)
  }
}

# now we sort them
rand_counts <- vapply(good_models7, count_random, integer(1))
AIC_vals <- sapply(good_models7, AIC)
nvar = order(-rand_counts)
aics = order(AIC_vals, decreasing=FALSE)

# now we take the sum of the position of each model
var_ix = match(1:length(good_models7), nvar)
aic_ix = match(1:length(good_models7), aics)
score <- var_ix + aic_ix

sorted_models7 <- good_models7[order(score)]

# we pick the best/largest one
start_model7 <- sorted_models7[[1]]
print(formula(start_model7))
```

```
## logRTresidual ~ agreement * surprisal + agreement * correct_number +
## (1 + agreement + surprisal | set_number) + (1 | UserId)
## <environment: 0x000001d0f42b2420>
```



```

# now we reduce the fixed effects - STEP keeps reducing random effects too
# so we go back to manual selection
fixed7 <- lapply(fixed_effects, fit_fixed, "(1 + agreement + surprisal | set_number) + (1 | U
serId)", subset(data, word_index==7))

# save the fixed models
saveRDS(fixed7, file='PATH/analysis/fixed7CAT.Rdata')

# now we check whether they converged
final_models7 <- list()
for (mdli in 1:length(fixed7)) {
  mdl <- fixed7[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code)))) {
    final_models7 <- append(final_models7, mdl)
  }
}

# there was only one model left (code not necessary)
fixed_aics = sapply(final_models7, AIC)
best_model7 = final_models7[order(fixed_aics, decreasing=FALSE)][[1]]
print(formula(best_model7))

```

```

## logRTresidual ~ surprisal + agreement + (1 + agreement + surprisal |
##   set_number) + (1 | UserId)
## <environment: 0x000001d0f374d2e8>

```

```
summary(best_model7)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: tmp_fm1a
## Data: data
## Control: lmerControl(optimizer = "nloptwrap", optCtrl = list(maxfun = 1e+06))
##
##      AIC      BIC    logLik deviance df.resid
##   -132.4   -64.8     77.2   -154.4     3447
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.3649 -0.5998 -0.0665  0.5157  4.3272
##
## Random effects:
## Groups      Name                Variance Std.Dev. Corr
## UserId      (Intercept)          0.002294 0.04790
## set_number   (Intercept)          0.002740 0.05234
##              agreementincorrect 0.002016 0.04490  -0.26
##              surprisalhigh       0.002000 0.04472  -0.41  0.20
## Residual                        0.052909 0.23002
## Number of obs: 3458, groups:  UserId, 88; set_number, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.03472    0.01184  56.01589  -2.932  0.00487 **
## surprisalhigh    0.02985    0.01055  38.80077   2.829  0.00736 **
## agreementincorrect 0.07029    0.01057  39.68508   6.649 6.01e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) srprsl
## surprisalhigh -0.436
## agrmntncrrc  -0.365  0.090
```

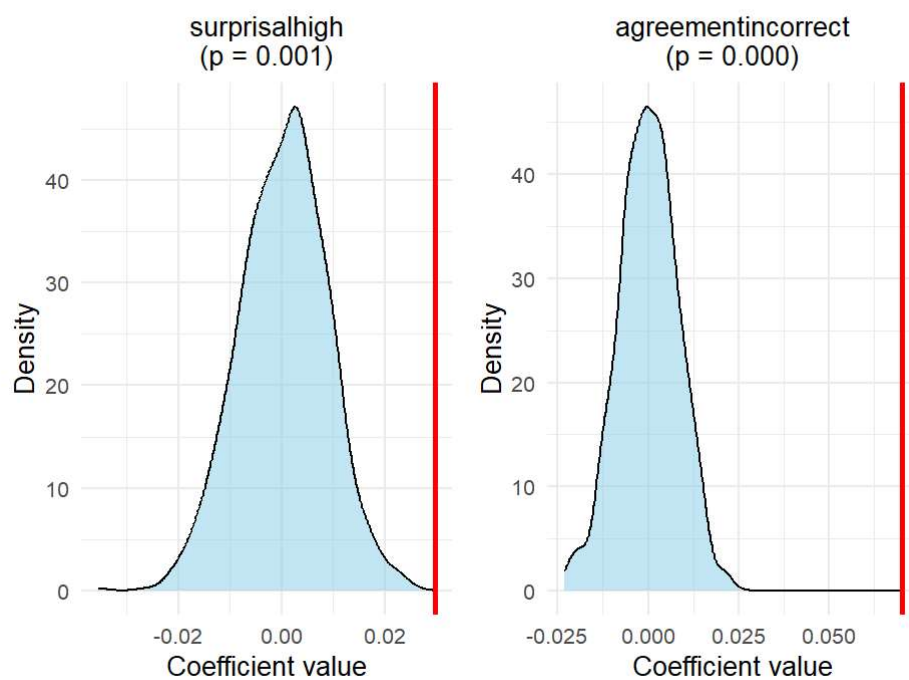
```
saveRDS(best_model7, file='PATH/analysis/best_model7CAT.Rdata')
```

Permutation test

```
best_model7 <- readRDS('PATH/analysis/best_model7CAT.Rdata')
perm7 <- permute_lmer(best_model7,
                     subset(data, word_index==7),
                     n_permutations,
                     verbose=TRUE)
saveRDS(perm7, file='PATH/analysis/perm7CAT.Rdata')
```

```
plots7 <- plot_permutation_densities(perm7)
print(plots7)
```

Permutation Test Distributions for Fixed Effects



```
print(perm7$p_values)
```

```
##      (Intercept)      surprisalhigh agreementincorrect  
##           0.000           0.001           0.000
```

Word index 8 (spill-over 3)

```
models8 <- readRDS('PATH/analysis/models8CAT.Rdata')

good_models8 <- list()
for (mdli in 1:length(models8)) {
  mdl <- models8[[mdli]]
  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code)))) {
    good_models8 <- append(good_models8, mdl)
  }
}

# now we sort them
rand_counts <- vapply(good_models8, count_random, integer(1))
AIC_vals <- sapply(good_models8, AIC)
nvar = order(-rand_counts)
aics = order(AIC_vals, decreasing=FALSE)

# now we take the sum of the position of each model
var_ix = match(1:length(good_models8), nvar)
aic_ix = match(1:length(good_models8), aics)
score <- var_ix + aic_ix

sorted_models8 <- good_models8[order(score)]

# we pick the best/largest one
start_model8 <- sorted_models8[[1]]
print(formula(start_model8))

## logRTresidual ~ agreement * surprisal + agreement * correct_number +
##      (1 + agreement + surprisal | set_number) + (1 + agreement |
##      UserId)
## <environment: 0x000001d0f8e45b20>
```

```

# now we reduce the fixed effects - STEP keeps reducing random effects too
# so we go back to manual selection
fixed8 <- lapply(fixed_effects, fit_fixed, "(1 + agreement + surprisal | set_number) + (1 + a
greement |
  UserId)", subset(data, word_index==8))

# save the fixed models
saveRDS(fixed8, file='PATH/analysis/fixed8CAT.Rdata')

# now we check whether they converged
final_models8 <- list()
for (mdli in 1:length(fixed8)) {

  mdl <- fixed8[[mdli]]

  if ((isSingular(mdl) == FALSE)&(mdl@optinfo$conv$opt == 0L)&(is.null(mdl@optinfo$conv$lme4
$code)))) {
    final_models8 <- append(final_models8, mdl)
  }
}

# they did, so we pick the model with the highest AIC
fixed_aics = sapply(final_models8, AIC)
best_model8 = final_models8[order(fixed_aics, decreasing=FALSE)][[1]]
print(formula(best_model8))

```

```

## logRTresidual ~ agreement * surprisal + (1 + agreement + surprisal |
##   set_number) + (1 + agreement | UserId)
## <environment: 0x000001d0f894cd48>

```

```
summary(best_model8)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: tmp_fm1a
## Data: data
## Control: lmerControl(optimizer = "nloptwrap", optCtrl = list(maxfun = 1e+06))
##
##      AIC      BIC    loglik deviance df.resid
##   -349.1   -263.0    188.6   -377.1     3463
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5939 -0.6114 -0.0605  0.5231  4.4538
##
## Random effects:
## Groups      Name                Variance Std.Dev. Corr
## UserId      (Intercept)          0.002248 0.04742
##              agreementincorrect 0.001780 0.04219  0.05
## set_number  (Intercept)          0.004260 0.06527
##              agreementincorrect 0.002023 0.04498 -0.31
##              surprisalhigh       0.001616 0.04021 -0.70  0.17
## Residual                        0.049052 0.22148
## Number of obs: 3477, groups:  UserId, 88; set_number, 40
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)      -0.01487    0.01372   59.71852  -1.084
## agreementincorrect    0.05751    0.01356   90.10812   4.241
## surprisalhigh       0.02549    0.01237   96.34349   2.061
## agreementincorrect:surprisalhigh -0.02622    0.01504 3198.13360  -1.743
##
##              Pr(>|t|)
## (Intercept)          0.2828
## agreementincorrect    5.38e-05 ***
## surprisalhigh        0.0420 *
## agreementincorrect:surprisalhigh 0.0815 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) agrmnt srprsl
## agrmntncrrc -0.418
## surprislhgh -0.602  0.383
## agrmntncrr: 0.273 -0.555 -0.605
```

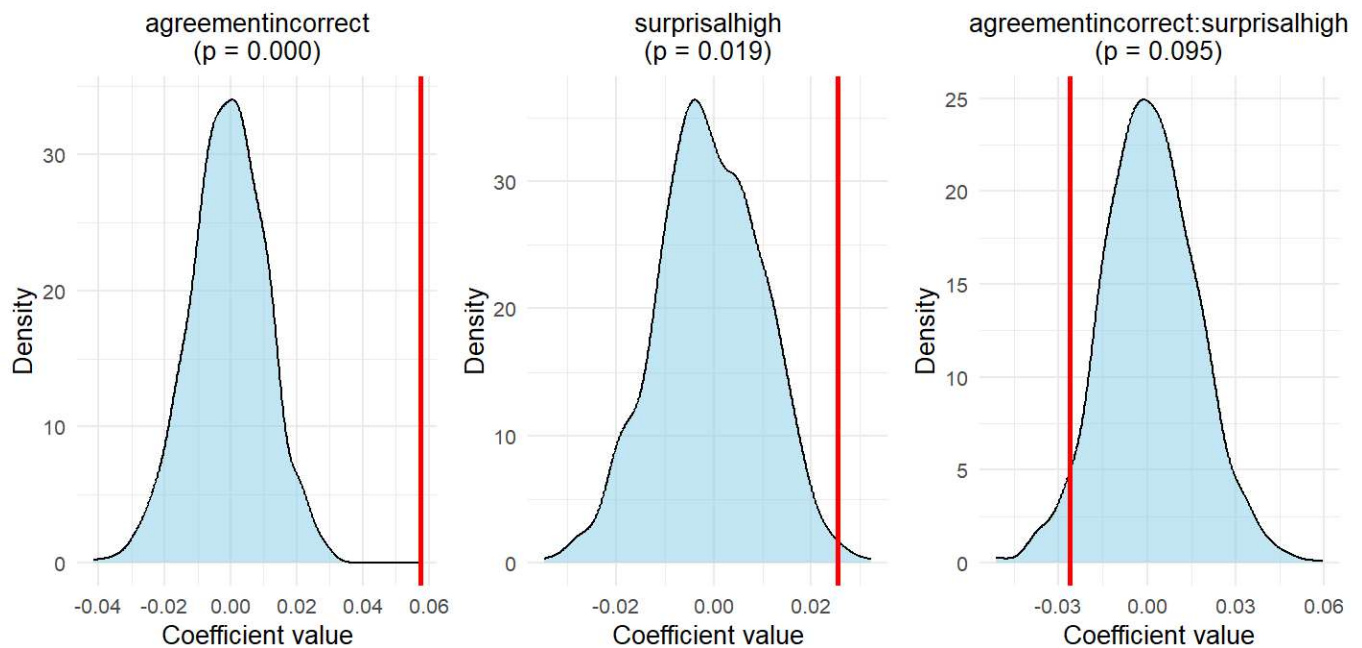
```
saveRDS(best_model8, file='PATH/analysis/best_model8CAT.Rdata')
```

Permutation test

```
best_model8 <- readRDS('PATH/analysis/best_model8CAT.Rdata')
perm8 <- permute_lmer(best_model8,
                      subset(data, word_index==8),
                      n_permutations,
                      verbose=TRUE)
saveRDS(perm8, file='PATH/analysis/perm8')
```

```
plots8 <- plot_permutation_densities(perm8)
print(plots8)
```

Permutation Test Distributions for Fixed Effects



```
print(perm8$p_values)
```

```
##          (Intercept)          agreementincorrect
##          0.784          0.000
##    surprisalhigh agreementincorrect:surprisalhigh
##          0.019          0.095
```